

# LCC Analysis K-1305 Sour oil leak

ORTL: RA-2025-312 Since K-1305 was restarted after MTA 2025, the operator has observed an unusually high seal oil consumption of approximately 150 L/day, compared to the normal rate of about 60 L/day. This excessive consumption could potentially lead to an unplanned shutdown for mechanical repairs.

|                             |                               |                    |                     |                                     |
|-----------------------------|-------------------------------|--------------------|---------------------|-------------------------------------|
| ASSET / EQUIPMENT<br>K-1305 | ANALYST<br>ENRM-RE-Thaninthon | DATE<br>2025-12-15 | LIFESPAN<br>6 Years | DISCOUNT RATE<br>5% (Inflation: 3%) |
|-----------------------------|-------------------------------|--------------------|---------------------|-------------------------------------|

## Analysis Results

Projection over 6 years

### Top Recommendations

#1

**Option 4 Rent a Regenerative installed on-site.** WINNER

Rent a Regenerative braking system and have it installed on-site.

TOTAL NPV

6,364,017

Best Choice

#2

**Option 3 Sent Sour to regen (Rayong)** LOWEST INITIAL COST

Transporting oil for regeneration in Rayong. (Environment not support)

TOTAL NPV

10,383,050

+4,019,033 vs #1

#3

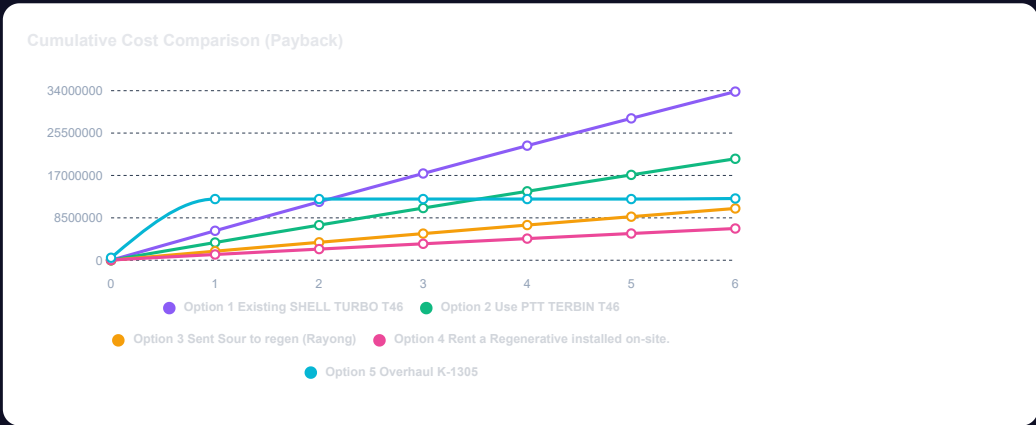
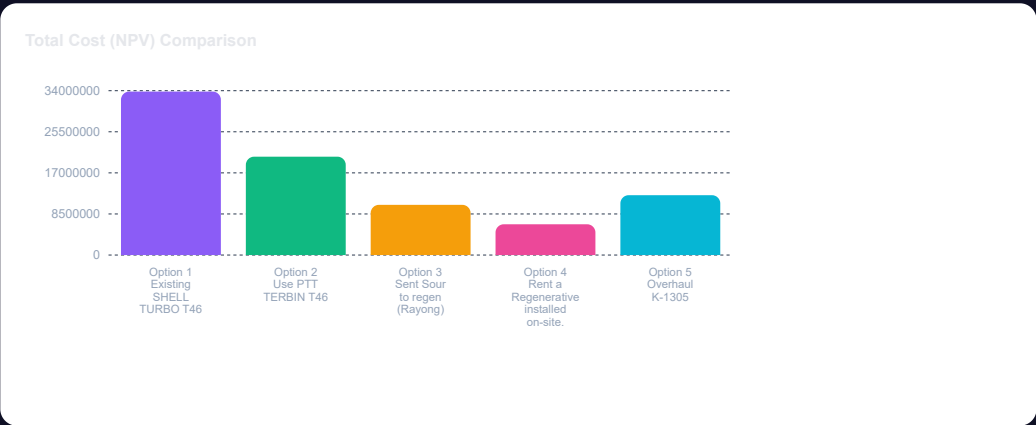
**Option 5 Overhaul K-1305**

Shutdown unit and overhaul K-1305

TOTAL NPV

12,378,351

+6,014,334 vs #1



Detailed Breakdown

| Metric                     | Option 1 Existing SHELL TURBO T46             | Option 2 Use PTT TERBIN T46 | Option 3 Sent Sour to regen (Rayong)                                      | BEST VALUE                                                        |  | Option 5 Overhaul K-1305          |
|----------------------------|-----------------------------------------------|-----------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------|--|-----------------------------------|
| Option Details             | Use SHELL TURBO T46 consumption rate 150 L/D. | Change to PTT TERBIN T46.   | Transporting oil for regeneration in Rayong.<br>(Environment not support) | Option 4 Rent a Regenerative installed on-site.                   |  | Shutdown unit and overhaul K-1305 |
|                            |                                               |                             |                                                                           | Rent a Regenerative braking system and have it installed on-site. |  |                                   |
| Initial Cost               | 0 Baht                                        | 0 Baht                      | 0 Baht                                                                    | 500,000 Baht                                                      |  | 500,000 Baht                      |
| Base Cost                  | 0 Baht                                        | 0 Baht                      | 0 Baht                                                                    | -                                                                 |  | -                                 |
| Installation               | -                                             | -                           | -                                                                         | 50,000 Baht                                                       |  | 500,000 Baht                      |
| Maintenance (NPV)          | 0 Baht                                        | 0 Baht                      | 9,232,496 Baht                                                            | 6,173,705 Baht                                                    |  | 0 Baht                            |
| Sent to regen (Rayong)     | -                                             | -                           | 9,092,184 Baht                                                            | -                                                                 |  | -                                 |
| Manhour                    | -                                             | -                           | 140,311 Baht                                                              | 0 Baht                                                            |  | 0 Baht                            |
| Rent a Regenerative        | -                                             | -                           | -                                                                         | 6,173,705 Baht                                                    |  | -                                 |
| Operation (NPV)            | 33,815,067 Baht                               | 20,345,165 Baht             | 1,150,554 Baht                                                            | 140,311 Baht                                                      |  | 0 Baht                            |
| SHELL TURBO T46 (4500 L/M) | 33,674,756 Baht                               | -                           | -                                                                         | -                                                                 |  | -                                 |
| Manhour                    | 140,311 Baht                                  | 140,311 Baht                | 140,311 Baht                                                              | -                                                                 |  | -                                 |
| PTT TERBIN T46 (4500 L/M)  | -                                             | 20,204,853 Baht             | -                                                                         | -                                                                 |  | -                                 |
| Add new oil                | -                                             | -                           | 1,010,243 Baht                                                            | -                                                                 |  | -                                 |
| Man hour                   | -                                             | -                           | -                                                                         | 140,311 Baht                                                      |  | -                                 |
| Failure & Loss (NPV)       | 0 Baht                                        | 0 Baht                      | 0 Baht                                                                    | 11,771,429 Baht                                                   |  | 11,771,429 Baht                   |
| New Item                   | -                                             | -                           | -                                                                         | -                                                                 |  | 11,771,429 Baht                   |
| Safety & Env (NPV)         | 0 Baht                                        | 0 Baht                      | 0 Baht                                                                    | 106,922 Baht                                                      |  | 0 Baht                            |
| End of Life (NPV)          | 0 Baht                                        | 0 Baht                      | 0 Baht                                                                    | 106,922 Baht                                                      |  | 106,922 Baht                      |
| Total NPV                  | 33,815,067 Baht                               | 20,345,165 Baht             | 10,383,050 Baht                                                           | 6,364,017 Baht                                                    |  | 12,378,351 Baht                   |
| Annualized Cost (EAC)      | 6,662,159 Baht                                | 4,008,353 Baht              | 2,045,642 Baht                                                            | 1,253,822 Baht                                                    |  | 2,438,751 Baht                    |

Key Metrics

**NPV (Net Present Value):** The total value of all future cash flows (costs) discounted to today's value. Lower is better.

**EAC (Equivalent Annual Cost):** The annualized cost of owning and operating the asset over its lifespan. Useful for comparing options with different lifespans.

Parameters

**LCC (Life Cycle Cost):** The total cost of ownership, including initial investment, operation, maintenance, and disposal costs, minus any residual value.

**Discount Rate:** The interest rate used to convert future costs into present value, reflecting the time value of money.

**Sensitivity Analysis:** Measures how changes in key assumptions (like discount rate or costs) affect the final NPV.

**Monte Carlo Sim:** Estimates the probability of different NPV outcomes by simulating range of input variations.